

9. Enzyme trypsin digests

- (a) proteins (b) carbohydrates
(c) fats (d) None of the above

B.P.S.C. Assistant Mains (GK) Paper II 2023

Ans. (a)

Enzyme Trypsin digests proteins. In the Small Intestine, Trypsin breaks down proteins, continuing the process of digestion that began in the stomach. Trypsin may also be referred to as a proteolytic enzyme or proteinase.

10. Glycogen stored in liver and muscles of human body is in form of-

- (a) Monosaccharide (b) Polysaccharide
(c) Protein (d) More than one of the above
(e) None of the above

BPSC (Tre-3) {Class 9-10}-2024

Ans. (b)

Glycogen is a multibranched polysaccharide of glucose that is stored in the liver and muscles of the human body. Glycogen is the primary way the body stores glucose a simple sugar that comes from Carbohydrates in food. When the body doesn't need Glucose for energy, it stores it as glycogen, when the body needs a quick energy boost, or when it's not getting glucose from food, glycogen is broken down to release glucose into the bloodstream.

III. Vitamins and Nutrition

1. The human body does not produce

- (a) DNA (b) vitamins
(c) hormones (d) enzymes
(e) None of the above / More than one of the above

B.P.S.C. Assistant Audit Officer 2022

Ans. (b)

Vitamins are essential substance that cannot be created by the human body vitamins are organic molecules that are essential to an organism in small quantities for proper metabolic function.

2. What is the chemical name of vitamin B₁₂?

- (a) Thiamin (b) Cobalamin
(c) Niacin (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift II

Ans. (b)

Vitamin B₁₂ also known as Cobalamin is a water-soluble vitamin that is mostly derived from the animal product such as meat, dairy and eggs.

3. The chemical name of Vitamin C is-

- (a) Ascorbic Acid (b) Thiamine
(c) Citric Acid (d) Tartaric acid

42nd B.P.S.C. (Pre) 1997

Ans. (a)

Indian gooseberry (amla) is the richest source of vitamin C (Ascorbic acid). Orange, tomato, lemon, papaya, peas, etc. are also good sources of vitamin C. It is an important antioxidant which helps to protect against cancer, heart disease, stress, maintaining a healthy immune system and wound healing process. It is essential for sperm production and formation of collagen. The collagen protein involved in the building and health of cartilage, joints, skin and blood vessels etc.

4. Which vitamin is found in citrus fruits and is essential in keeping the skin healthy -

- (a) Vitamin 'A' (b) Vitamin 'B'
(c) Vitamin 'C' (d) Vitamin 'D'

44th B.P.S.C. (Pre) 2000-01

Ans. (c)

Vitamin C is found in juicy fruits, especially amla, lemon, orange, guava and it is necessary to keep the religion healthy. The chemical name of vitamin 'C' is ascorbic acid.

5. The vitamin which is effective in blood clotting is :

- (a) Vitamin A (b) Vitamin B
(c) Vitamin D (d) Vitamin K
(e) None of the above / More than one of the above

66th B.P.S.C. (Pre) 2020

Ans. (d)

Vitamin K is a responsible factor for blood clotting or coagulation. Without vitamin K, blood coagulation is seriously impaired and uncontrolled bleeding occurs. Vitamin K is known as Antihemorrhagic factor.

6. Which of the following vitamins is used as an antidote to anticoagulant poisons?

- (a) Vitamin A (b) Vitamin D
(c) Vitamin E (d) Vitamin K
(e) None of the above/More than one of the above

60th to 62nd B.P.S.C. (Pre) 2016

Ans. (d)

Vitamin 'K' is used as an antidote to anticoagulant poisons. This vitamin is called antihemorrhagic factor. Operation of persons with deficiency of vitamin K can not be done easily because there is a fear of excess bleeding.

7. Which of the following vitamins is known as Ergocalciferol?

- (a) Vitamin D₂ (b) Vitamin D₃
(c) Vitamin B₁₂ (d) Vitamin B₆
(e) None of the above/More than one of the above

60th to 62nd B.P.S.C. (Pre) 2016

Ans. (a)

There are two major chemical forms of vitamin D, namely vitamin D₂ sometimes referred to as Ergocalciferol and vitamin D₃ sometimes referred to as Cholecalciferol. Vitamin C is known as ascorbic acid whereas vitamin A is known as retinol. Folic acid is the name of Vitamin B₉.

8. In human body, Vitamin A is stored in :

- (a) Liver (b) Skin
(c) Lung (d) Kidney
(e) None of the above/More than one of the above

60th to 62nd B.P.S.C. (Pre) 2016

Ans. (a)

Vitamin A is largely stored in the liver in the human body. Its chemical name is Retinol whose deficiency causes night blindness.

9. Which of the following vitamins contain Cobalt –

- (a) Vitamin B₆ (b) Vitamin B₂
(c) Vitamin B₁ (d) Vitamin B₁₂

48th to 52nd B.P.S.C. (Pre) 2008

Ans. (d)

Vitamin B₁₂ (Cobalamin) is a water-soluble vitamin. It contains a metallic ion Cobalt. Its chemical formula is C₆₃H₈₈CoN₁₄O₁₄P. It functions as a co-factor for enzymes in the metabolism of amino acids and fatty acids, required for new cell synthesis, normal blood formation and neurological function. Its deficiency causes pernicious anemia, nervous system abnormality etc.

10. Which of the following pairs is not correctly matched?

- (a) Vitamin B₁ - Oranges
(b) Vitamin D - Cod-liver oil
(c) Vitamin E - Wheat germ oil
(d) Vitamin K - Alfalfa
(e) None of the above/More than one of the above

60th to 62nd B.P.S.C. (Pre) 2016

Ans. (e)

Sunlight and cod-liver oil are rich source of vitamin D. Wheat germ oil is rich in vitamin E, while alfalfa is rich in vitamin K. Though orange is rich in vitamin C but it is also a low source of vitamin B₁. Thus option (e) is the most probable answer.

11. Which of the following pairs is not correctly matched?

- (a) Retinol - Xerophthalmia
(b) Tocopherol - Beri-beri
(c) Cyanocobalamin - Anaemia
(d) Ergocalciferol - Rickets
(e) None of the above/More than one of the above

60th to 62nd B.P.S.C. (Pre) 2016

Ans. (b)

Deficiency of retinol i.e. vitamin A leads to Xerophthalmia. Lack of cyanocobalamin (vitamin B₁₂) causes pernicious anaemia and lack of vitamin D (Cholecalciferol 'D₃' and Ergocalciferol 'D₂') causes rickets in children. Beri-beri is caused by lack of vitamin 'B₁' i.e. thiamin not due to lack of tocopherol (vitamin 'E').

12. Which of the following is the best source of vitamin A?

- (a) Wheat (b) Honey
(c) Carrot (d) Peanut
(e) None of the above / More than one of the above

B.P.S.C. CDPO 2022

Ans. (c)

A carrot's main nutrient, beta-carotene (responsible for this root vegetable's characteristic orange color), is a precursor to vitamin A and helps your eyes to adjust in dim conditions. Night blindness is one of the first signs of vitamin A deficiency.

13. Legumes are highly nutritious because they are rich in :

- (a) fat (b) protein
(c) oil (d) starch
(e) None of the above / More than one of the above

66th B.P.S.C. (Pre) 2020

Ans. (b)

Legumes are highly nutritious because they are rich in proteins. Legumes are typically low in fat, contain no cholesterol, and are high in folate, potassium, iron and magnesium.

14. Branching series in a food chain is called

- (a) food system (b) food level
(c) food web (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (c)

A food web consists of all the food chains in a single ecosystem. Each living thing in an ecosystem is part of multiple food chains. Each food chain is one possible path that energy and nutrients may take as they move through the ecosystem.

15. _____ are present in the first trophic level in the food chain.

- (a) Autotrophs (b) Primary consumers
(c) Tertiary consumers (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (a)

An autotroph is an organism that can produce its own food using light, water, carbon dioxide, or other chemicals. Because autotrophs produce their own food, they are sometimes called producers. autotrophs are present in the first trophic level in the food chain.

16. Which of the following constitute a food chain?

- (a) Grass, deer and tiger
(b) Grass, fish and cow
(c) Grass, rabbit and elephant
(d) More than one of the above
(e) None of the above

BPSC (Tre-3) {Class 1-5}–2024

Ans. (a)

Food chain, in ecology, is the sequence of transfers of matter and energy in the form of food from organism to organism. Food chains intertwine locally into a food web because most organisms consume more than one type of animal or plant. The correct order of the food chain in the given question– Grass → Deer → Tiger

17. In a food chain, which trophic level has highest energy level?

- (a) Consumers of first trophic level
(b) Consumers of top trophic level
(c) Producers
(d) More than one of the above
(e) None of the above

BPSC (Tre-3) {Class 1-5}–2024

Ans. (c)

The first trophic level consists of producers, which are organisms capable of producing their own food through processes like photosynthesis. In photosynthesis, for example, plants, algae, and some bacteria (like cyanobacteria) convert sunlight, water, and carbon dioxide into glucose (a form of sugar) and oxygen. Since they make their own food, they are called autotrophs.

This is a key aspect of ecosystems because autotrophs serve as the foundation of the food chain, providing energy for organisms at higher trophic levels. Heterotrophs, which include herbivores, carnivores, and omnivores, rely on the energy produced by autotrophs to survive.

18. Which of the food grains is richest in protein?

- (a) Gram (b) Pea
(c) Soyabean (d) Wheat

40th B.P.S.C. (Pre) 1995

Ans. (c)

Soyabean is the richest source of protein. It contains about 40% protein. It is classified as an oilseed as well as pulse crop. The milk is produced by soyabean which is equivalent to the milk of a cow. Each grain of soyabean contains about 40% protein, 20-23% fat and 33.5% carbohydrate.

19. Which of the following is mostly carbohydrate?

- (a) Wheat (b) Barley
(c) Rice (d) Maize
(e) None of the above/More than one of the above

63rd B.P.S.C. (Pre) 2017

Ans. (c)

Among the given options, rice has the highest percentage of carbohydrates.

20. The ultimate substance to which the carbohydrates are degraded, is –

- (a) amino acid (b) glycerol
(c) glucose (d) maltose
(e) None of the above/More than one of the above

67th B.P.S.C. (Re-Exam) (Pre) 2021

Ans. (c)

After the breakdown of carbohydrates in digestion glucose is formed which provides energy to cells.

21. Milk is a poor source of :

- (a) calcium (b) protein
(c) vitamin C (d) carbohydrate
(e) None of the above / More than one of the above

66th B.P.S.C. (Pre) 2020

Ans. (c)

Milk is considered a complete food because it provides protein, vitamin A, vitamin B₁, B₂, B₆, B₁₂, carbohydrate, calcium and potassium etc. Milk contains very small amounts of vitamin C. Hence, option (c) is the correct answer.

22. What is triglyceride?

- (a) Protein (b) Carbohydrate
(c) Fat (d) Mineral
(e) None of the above/More than one of the above

64th B.P.S.C. (Pre) 2018

Ans. (c)

A triglyceride is an ester derived from glycerol and three fatty acids. Triglycerides are the main constituents of body fat in humans and other vertebrates, as well as vegetable fat. The high level of triglycerides is responsible for obesity and high blood pressure.